# How Much Targeted Training Does It Take to Top a Multi-Task Leaderboard? Minimum Investment Under Different Aggregation Rules

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## Abstract

A leaderboard turns per-task scores into one rank, so a developer can try to lift a chosen model by training on data related to individual sub-datasets. How much such targeted training is needed to put the model first, and how does that depend on the aggregation rule? We define the investment as the minimum raise of the target's task scores, within a per-task cap, that makes it strictly first, and measure it in points, tasks touched and largest single-task jump. We solve it exactly for the mean, median, minimum, Borda and Condorcet-winner rules, prove closed forms for the first three, and apply it to every model of the Open LLM Leaderboard (4495 targets, 6 tasks) and of HELM MMLU (78 targets, 57 subjects) under the score rules; Borda and Condorcet are solved on OLLB for the 50 best targets only, and Copeland on HELM only. With caps taken from observed fine-tuning gains, first place is reachable for only 12% of OLLB targets under the mean (largest feasible rank 901) and 19% of HELM targets (largest feasible rank 16); with a 100-point cap every target can reach it. Among feasible targets the mean is the costliest rule and the median and minimum the cheapest: at OLLB ranks 11–50 the other rules need 0.18–0.50 of the mean's total points. Conclusions depend on the cap, which on HELM is transferred from OLLB, and hold for fixed rivals and score tables only.

## 1 Introduction

Multi-task benchmarks rank models by aggregating per-task scores into one number, and the resulting rank influences which models are adopted, published and funded. A developer therefore has an incentive to improve a chosen target model specifically on the benchmark's sub-datasets, for instance by training on data related to them. How much such targeted improvement is needed to reach first place, and how that amount depends on the aggregation rule, is the question we study on public leaderboard data.

Our approach is deliberately simple and exact. We take the score table of a public leaderboard as fixed, let the target raise its own scores task by task up to a cap, and compute the *minimum* raise that makes it strictly first under a given aggregation rule. The cap is the central modelling assumption: it stands for how far training on related data can move one score. We therefore report results under a data-grounded cap and an oracle cap, and report infeasibility (no allowed raise makes the target first) as an outcome rather than a missing value.

**Contributions.** (i) A formulation of the minimum targeted investment with three cost measures ($L_1$, $K$, $L_\infty$) and an explicit family of caps (Section 3). (ii) Exact solvers for score rules (mean,

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

median, minimum, power and trimmed means) and rank rules (Borda, Condorcet winner, plurality, Copeland), tested against brute force, with closed forms and proofs for the mean, minimum and median (Section 3.4, Appendix A). (iii) A study of every model on two public leaderboards (rank rules on OLLB: the 50 best targets), with bootstrap and evaluation-noise intervals (Sections 4–5).

**Findings.** Under caps grounded in observed fine-tuning gains, first place is out of reach for most models, under every rule: the feasible fraction is 12% (mean) on OLLB and 19% on HELM, and the rule changes how far down the board it is reachable. Where it is reachable, the mean rule costs the most points, and order-statistic and rank rules cost between 18% and 50% of that at OLLB ranks 11–50. Two of three hypotheses fixed beforehand are only partly borne out (Section 5). All conclusions are conditional on the cap and on the other models being fixed; Section 6 lists where they fail.

## 2 Related work

**Benchmarks and leaderboards.** MMLU [Hendrycks et al., 2020] aggregates 57 subjects, and HELM [Liang et al., 2022] releases per-scenario scores and raw model outputs for many models; BIG-Bench Hard [Suzgun et al., 2022] selects hard tasks from BIG-Bench. These provide the multi-task score matrices that we analyse.

**Fragility of rankings.** Dehghani et al. [2021] show that the relative order of methods can change with the choice of benchmark tasks (the "benchmark lottery"). Singh et al. [2025] document how private testing and unequal data access distort the Chatbot Arena leaderboard. Both concern what is measured and disclosed; we instead hold the leaderboard fixed and ask what a single developer must change for one model to win.

**Aggregation and social choice.** Lanctot et al. [2023] treat each task as a voter and use voting theory to build cross-task evaluation; the Handbook of Computational Social Choice [Brandt et al., 2016] covers the rules (Borda, Condorcet-consistent rules, plurality) we reuse. Our rules are the same families, but our question is the cost of changing the winner, not the choice of rule.

**Training on the test task and contamination.** Zhou et al. [2023] show that benchmark leakage into training can raise scores sharply, and Yang et al. [2023] show that rephrased test samples evade common decontamination. Chen et al. [2026] analyse, in a game-theoretic model, the incentives that benchmarks create for strategic post-training. Our analysis is empirical and combinatorial: given a fixed score table, we compute the exact minimum targeted improvement per rule, under explicit assumptions about how much any one task can be improved.

**Positioning.** Closest in question is Chen et al. [2026], who model developers choosing how much to spend on benchmark-specific improvement of a single benchmark score, prove non-existence of equilibria for current protocols and a unique equilibrium for tune-before-test. That analysis concerns one score per model; the choice of aggregation rule over tasks does not enter it. We take the opposite stance: no game and no equilibrium, a fixed table, and the exact cost for one developer under each of several aggregation rules. Colombo et al. [2022] propose a social-choice-based aggregation for NLP benchmarks and show it changes conclusions relative to the mean; we do not propose a rule but measure how costly it is to move a model to first place under each. The computational-social-choice literature studies the analogous *bribery* problem, the cost of changing votes so that a chosen candidate wins under a voting rule [Faliszewski et al., 2009, Brandt et al., 2016]. Our setting differs in that the manipulator changes its own scores on tasks (the "voters") and pays in score points under per-task caps, which gives closed forms for score rules and integer programs for rank rules.

## 3 Problem setup

### 3.1 Leaderboard, target and investment

A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$ of $M$ models on $n$ tasks. Fix a target model $i$ with score vector $x = X_{i\cdot}$; all other rows stay fixed. In targeted training the developer can raise the

target's score on task $t$ to any $y_t \in [x_t, c_t]$, where $c_t$ is a *cap* (Section 3.2). An aggregation rule $f$ maps the score vectors to a ranking; the target is *first* if it is strictly above every other model under $f$ (by a margin $\varepsilon = 10^{-3}$ points for score rules). We report the investment in three ways:

$$L_1 = \sum_t (y_t - x_t), \qquad K = \#\{t : y_t > x_t\}, \qquad L_\infty = \max_t (y_t - x_t),$$

the total points, the number of tasks touched, and the largest single-task jump. The minimum of each is taken separately; $L_\infty$ is also the cost of the uniform "general improvement" that raises every task by the same amount. For comparison across rules and datasets we also report $K/n$ and $L_1/n$.

### 3.2 Caps: how much can one task be improved?

Without a bound on $y_t$ the question is trivial (put every task at 100). The cap $c_t$ is therefore the central modelling assumption, and we never collapse it to a single choice. Two caps appear in the main text: `ceil`, $c_t = 100$, an oracle upper bound on what can be done and so a lower bound on any required investment, far above any observed score (on OLLB the best GPQA score is 29.4 and the best MuSR score 38.7, and scores are floored at 0), so `ceil` is to be read as an oracle and not as a plausible target; and `emp99`, $c_t = \min(100, x_t + g_t^{99})$ where $g_t^{99}$ is the 99th percentile of the observed gains (fine-tune minus its declared base model) on task $t$ in the Open LLM Leaderboard. `emp99` is the only data-grounded cap. It inherits a selection bias, since only published fine-tunes appear, and it measures gains attained by the community, not the gains a determined developer could attain. Further caps (best existing score $+0$ or $+2$, and at most $+5/+10/+20$ points per task) are reported in the appendix. When no $y \le c$ makes the target first, the case is *infeasible*; infeasibility is reported as a rate and never coded as a large investment, and all investment statistics are conditional on feasibility.

### 3.3 Aggregation rules

Score rules apply a function $f$ to the score vector of each model: the *mean* (the rule of both leaderboards we use), the *median*, and the *minimum* (worst-task score). Rank rules use only the order of models on each task: *Borda* (sum over tasks of the number of models beaten, ties counting one half; equivalent to mean rank; on HELM this is "Borda (mean rank) on the 57 subjects", which we compute ourselves and which is not HELM's own published ranking, since HELM's mean win rate is not in the files we downloaded) and the *Condorcet-winner* requirement (the target beats every rival on a strict majority of tasks, ties counting one half). Being a Condorcet winner is sufficient for being first under any Condorcet-consistent rule such as Copeland or Schulze, so its investment upper-bounds theirs; for Copeland we also solve the problem exactly on HELM. Geometric, harmonic and power means, the trimmed mean and plurality are in the appendix.

### 3.4 Exact minimum-investment solvers

For a score rule let $\tau = \max_{j \ne i} f(X_{j\cdot})$ be the level of the best rival; the target is first iff $f(y) \ge \tau + \varepsilon$. For the mean, the minimum and the median the minimum investment has a closed form (Propositions 1–3, proved in Appendix A); for power means and the trimmed mean we solve a convex program or enumerate exactly. For rank rules a raise of task $t$ to level $s + \varepsilon$ passes exactly the models with score at most $s$ on that task, so only these finitely many levels need be considered, and we solve the resulting integer program exactly with HiGHS, adding rival constraints lazily. Every solver is checked against brute-force search on small random instances (unit tests in the supplementary code). For each target we report $L_1$ and $K$, each minimised separately, and $L_\infty$ for the uniform baseline.

**Proposition 1** (Mean). *Let $f$ be the mean and $h_t = c_t - x_t$. The target can become first iff $\sum_t c_t > n\tau$ (up to $\varepsilon$). The minimum $L_1$ is $n(\tau - f(x))$ regardless of how it is spread, the minimum $K$ is the least $k$ for which the $k$ largest $h_t$ sum to at least this amount, and the minimum $L_\infty$ is $\tau - f(x)$ (raise all tasks equally) when no cap binds.*

**Proposition 2** (Minimum). *Let $f = \min_t y_t$. The target can become first iff $c_t > \tau$ for all $t$ with $x_t \le \tau$. Then $L_1 = \sum_t (\tau - x_t)^+$ and $K = \#\{t : x_t \le \tau\}$: every deficient task must be raised.*

**Proposition 3** (Median, $n$ odd). *Let $m = (n+1)/2$, $d_t = (\tau - x_t)^+$ for tasks with $c_t > \tau$ and $d_t = \infty$ otherwise. The target is first iff at least $m$ tasks end above $\tau$. The minimum $L_1$ is the sum of the $(m - \#\{t : x_t > \tau\})^+$ smallest finite $d_t$ among tasks with $x_t \le \tau$, and $K$ equals that count; the target can be made first iff enough finite $d_t$ exist.*

(All statements are for $\varepsilon \to 0$; the code uses $\varepsilon = 10^{-3}$ points. For even $n$ the median is the average of the two middle order statistics and we only have the computed solution.) We verify Propositions 1–3 numerically against the solvers (Section 4); no theorem is claimed for the rank rules.

# 4 Experimental setup

**Data.** We use two public leaderboards, each at a fixed, recorded version (checksums in the supplementary reproduction notes). *HELM MMLU*, release v1.13.0 (summary dated 2025-01-10): the exact-match score of 79 models on each of the 57 MMLU subjects ($79 \times 57$, no missing cell). *Open LLM Leaderboard* (OLLB) `contents`, dataset commit `9c09a7c` (2025-03-20): 4576 rows, of which 4497 have distinct model names after we keep the best-scoring variant of each duplicated name, and 4496 after removing one flagged model; each model has six normalised task scores (IFEval, BBH, MATH Lvl 5, GPQA, MUSR, MMLU-PRO), so $n = 6$. Merged models are kept in the main analysis and dropped in a robustness slice. The mean we use equals the sites' own aggregate (OLLB `Average`, HELM "MMLU All Subjects") to within $6 \times 10^{-14}$ points on every model. The mean is the official rule on both leaderboards; the median, minimum, Borda, Condorcet, plurality and Copeland rules are ours and are not published rankings. The 79 HELM models include closed-weight frontier models (for example GPT-4o, Claude 3.5, Gemini 1.5 and 2.0), which no outside developer can train; treating every model as a target that its developer can lift is a modelling assumption. Neither source declares a licence for the score data; we redistribute only the derived score tables.

**Targets, caps and what is reported.** A target is a model other than the leader under the mean (4495 targets on OLLB, 78 on HELM), taken in turn with all others fixed; all tables and figures use this same set for every rule, stratified by rank under the mean. "First" means strictly first by the margin $\varepsilon = 10^{-3}$, so a model that is first only through an exact tie is itself a target at cost $\varepsilon$. The solver tables therefore hold one more row for the HELM median (79 targets: Claude 3.5 Sonnet (20241022), the mean leader, and DeepSeek v3 have the identical median, so the mean leader costs $\varepsilon$ there); that row is a tie-break artefact and is left out of the analysis. On OLLB the primary cap is `emp99` (per-task 99th percentile of the gain of a fine-tune over its declared base model, estimated from 540 such pairs in the same leaderboard). HELM has no base/fine-tune pairs, so its primary cap, `emp99T`, applies the OLLB MMLU-PRO 99th-percentile gain to all 57 subjects; this is a transfer assumption, not a measurement on HELM. The oracle cap `ceil` is reported beside it. All investment statistics are medians over feasible targets, shown together with the feasible fraction. OLLB rank rules are exact integer programs, which at $M = 4496$ models we run only for the 50 best targets under the mean; we apply a 120 s limit per instance and report time-outs as censored, never as infeasible.

# 5 Results

All statements below are about targets that can be made first within the cap; the feasible fraction is shown beside every cost, and a rule is called cheaper than another only on targets that are feasible under both.

**The closed forms are exact.** Propositions 1–3 agree with the exact solvers on every target of both datasets and every cap: the largest deviation in $L_1$ is below $2 \times 10^{-12}$ points, the deviation in $K$ is zero, and no feasibility verdict differs (Appendix Table 3). The OLLB median is solver-only because $n = 6$ is even.

**Feasibility depends on rank and on the rule.** Under the data-grounded caps, targeted training cannot make most models first (Figure 1, left). On OLLB the feasible fraction under the mean and the median is close to one for targets up to rank a few hundred and near zero beyond rank about one thousand; on HELM every target at ranks 2–10 is feasible under the mean, median, Borda and Condorcet rules, and the fraction falls steeply afterwards (Table 2: for ranks 11–50 only 6 of 40 targets are feasible under the mean and 17 of 40 under the median). The minimum rule is the most restrictive on both datasets, since every task below the best rival's weakest score must be raised and any one cap can block it (Proposition 2). The cap, not the rule, decides feasibility far from the top: with the oracle cap `ceil` every target is feasible (Appendix Figure 4). Appendix Figures 5 and 6 vary the cap on the two datasets: the feasible share rises with the cap in every stratum, while the cost among feasible targets under the mean does not change.

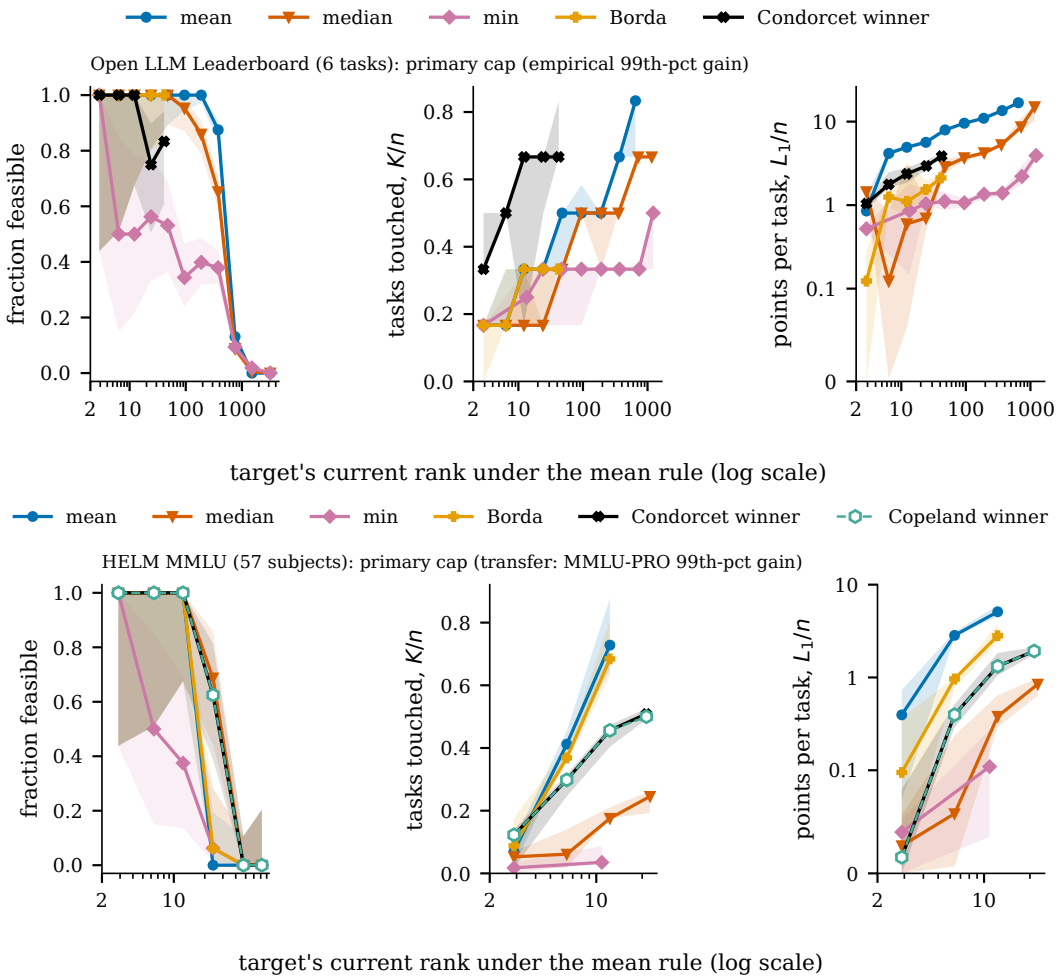



Figure 1: Minimum targeted investment to become first versus the target's current rank under the mean, primary caps. Left: fraction of targets that can reach first at all. Middle and right: tasks touched ($K/n$) and total points per task ($L_1/n$), *among feasible targets*. Top: OLLB, $n = 6$, cap `emp99` (4495 targets; Borda and Condorcet only for the 50 best). Bottom: HELM MMLU, $n = 57$, cap `emp99T` (78 targets; the cap is transferred from OLLB). Points are bin medians (bins double from rank 2; bins with fewer than 3 targets omitted), bands are 95% bootstrap intervals over targets. Generated by `figures/src/fig_rank_rules.py`.

Table 1 gives the frontier. On OLLB the fraction of targets that can reach first place under `emp99` is 12% under the mean, 10% under the median and 6% under the minimum; every target is feasible only up to rank 394 (mean), 90 (median) and 5 (minimum), and the largest feasible rank is 901, 1207 and 2225. On HELM under `emp99T` the fractions are 19%, 33% and 12%, every target is feasible up to rank 16, 19 and 6, and the largest feasible rank is 16, 29 and 20; the Condorcet requirement reaches rank 26 and Borda 17. The median is thus more exposed to targeted training than the mean on both datasets, and the minimum is feasible rarely but, when feasible, cheaply. The largest feasible rank can be an isolated case (OLLB, minimum), which is why the column "all feasible up to" is shown.

**Cost, conditional on feasibility, differs strongly by rule.** Among feasible targets, the mean rule is the most expensive on both datasets and the minimum and median rules the cheapest (Figure 1, right; Table 2). On OLLB at ranks 11–50 the median feasible target needs 37.8 points in total under the mean, 9.1 under the median, 6.9 under the minimum, 11.7 under Borda and 20.5 under the Condorcet-winner requirement; relative to the mean on jointly feasible targets, the ratios are 0.22, 0.18, 0.27 and 0.50. On HELM at ranks 11–50 the corresponding totals are 321, 38, 10, 188 and 105 points. The reason is structural: the mean must close the whole summed gap ($n$ times the gap,

Table 1: Feasibility frontier under the primary cap (`emp99` on OLLB, `emp99T` on HELM) and the oracle cap (`ceil`). *share*: fraction of targets for which first place is reachable (Wilson 95% intervals); $r^*$: largest rank under the mean among feasible targets (may be an isolated case); *all feas. up to*: largest $r$ such that every target of rank $\leq r$ is feasible; *first bin < 1/2*: first rank bin (bins double from rank 2) in which fewer than half are feasible. †: the 50 best targets under the mean only, not comparable with the score rules' fractions. Generated by `figures/src/tables.py`.

| cap | rule | targets | feasible | share [95% CI] | $r^*$ | all feas. up to | first bin < 1/2 |
|---|---|---|---|---|---|---|---|
| primary (`emp99`) | mean | 4495 | 546 | 0.12 [0.11, 0.13] | 901 | 394 | 513–1024 |
| primary (`emp99`) | median | 4495 | 451 | 0.10 [0.09, 0.11] | 1207 | 90 | 513–1024 |
| primary (`emp99`) | min | 4495 | 273 | 0.06 [0.05, 0.07] | 2225 | 5 | 65–128 |
| primary (`emp99`) | Borda† | 49 | 49 | 1.00 [0.93, 1.00] | 50 | 50 | none |
| primary (`emp99`) | Condorcet† | 49 | 42 | 0.86 [0.73, 0.93] | 50 | 18 | none |
| oracle (`ceil`) | mean | 4495 | 4495 | 1.00 [1.00, 1.00] | 4496 | 4496 | none |
| oracle (`ceil`) | median | 4495 | 4495 | 1.00 [1.00, 1.00] | 4496 | 4496 | none |
| oracle (`ceil`) | min | 4495 | 4495 | 1.00 [1.00, 1.00] | 4496 | 4496 | none |
| oracle (`ceil`) | Borda† | 49 | 49 | 1.00 [0.93, 1.00] | 50 | 50 | none |
| oracle (`ceil`) | Condorcet† | 49 | 49 | 1.00 [0.93, 1.00] | 50 | 50 | none |

| cap | rule | targets | feasible | share [95% CI] | $r^*$ | all feas. up to | first bin < 1/2 |
|---|---|---|---|---|---|---|---|
| primary (`emp99T`) | mean | 78 | 15 | 0.19 [0.12, 0.29] | 16 | 16 | 17–32 |
| primary (`emp99T`) | median | 78 | 26 | 0.33 [0.24, 0.44] | 29 | 19 | 33–64 |
| primary (`emp99T`) | min | 78 | 9 | 0.12 [0.06, 0.20] | 20 | 6 | 9–16 |
| primary (`emp99T`) | Borda | 78 | 16 | 0.21 [0.13, 0.31] | 17 | 17 | 17–32 |
| primary (`emp99T`) | Condorcet | 78 | 25 | 0.32 [0.23, 0.43] | 26 | 26 | 33–64 |
| primary (`emp99T`) | Copeland | 78 | 25 | 0.32 [0.23, 0.43] | 26 | 26 | 33–64 |
| oracle (`ceil`) | mean | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | median | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | min | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | Borda | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | Condorcet | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |
| oracle (`ceil`) | Copeland | 78 | 78 | 1.00 [0.95, 1.00] | 79 | 79 | none |

Proposition 1), whereas order-statistic rules pay only for the cheapest majority of tasks or the weakest tasks. Rank rules sit in between, because they pay for passing rivals task by task. Table 2 shows intervals; the ordering mean $>$ Condorcet $>$ Borda $\gtrsim$ median $\gtrsim$ min on OLLB is not uniform across rank bins (the intervals at ranks 2–10 are wide, with $n = 9$ targets).

Appendix Table 4 gives the paired comparison that supports a statement "rule A is cheaper than rule B": the median per-target ratio of the cost under the rule to that under the mean, on targets feasible under both, with the number of such targets out of those feasible under either. On OLLB the ratios over all jointly feasible targets are $0.39$ (median), $0.11$ (minimum), $0.27$ (Borda), $0.52$ (Condorcet) and $0.66$ (plurality); on HELM they are $0.07$, $0.05$, $0.51$, $0.21$ (Condorcet and Copeland) and $0.05$ for plurality. The comparison is selective by construction: the minimum rule is jointly feasible with the mean on only 237 of 581 OLLB targets, and for the 4 targets at ranks 2–5 on OLLB the median is not cheaper than the mean (ratio $1.38$, interval $[0.03, 2.00]$). The statement that the other rules are cheaper than the mean therefore holds on the targets where both can win, and is not a statement about all targets.

**Sensitivity to the setup.** Appendix Table 5 repeats the main comparison with the margin that defines "first" raised to 1 and 2 points, without merged models, with raw accuracies instead of normalised scores, and with random task weights. A larger margin lowers the feasible share and raises the cost by a factor of $1.1$–$1.6$ at one point and $1.2$–$2.3$ at two points, without changing the ordering of the rules in cost (HELM, margin 2: $270$, $39$ and $6.7$ points for the mean, median and minimum). Dropping merged models changes nothing for the mean and median (ratio $1.00$). With raw accuracies and a $+10$ point cap the median rule is almost never feasible on OLLB (below $1\%$ of targets, against $5\%$ with normalised scores), so the claim that the median reaches further down than the mean does not carry over to that setting. Random task weights lower the cost of the weighted mean (median ratio to the uniform mean $0.43$–$0.70$ on jointly feasible draws), so the cost we report

Table 2: Median investment to become first among feasible targets, by rank stratum and rule, primary caps: OLLB (`emp99`, top) and HELM (`emp99T`, bottom). "feas." = feasible targets / targets in the stratum; $K$, $L_1$ with 95% bootstrap intervals over targets; the last column is the median ratio of $L_1$ to the mean rule over targets feasible under both. OLLB Borda and Condorcet use the 50 best targets only. Generated by `figures/src/tables.py`.

| | rank 2–10 | | | | rank 11–50 | | | |
|---|---|---|---|---|---|---|---|---|
| | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ |
| mean | 9/9 | 1.0 [1.0, 2.0] | 24.6 [5.1, 27.7] | – | 40/40 | 2.0 [2.0, 2.0] | 37.8 [33.9, 41.0] | – |
| median | 9/9 | 1.0 [1.0, 1.0] | 0.8 [0.0, 8.4] | 0.03 [0.00, 1.70] (9) | 40/40 | 1.0 [1.0, 2.0] | 9.1 [4.4, 14.6] | 0.22 [0.13, 0.40] (40) |
| min | 5/9 | 1.0 [1.0, 2.0] | 3.7 [2.8, 14.6] | 0.61 [0.23, 0.77] (5) | 24/40 | 2.0 [1.5, 2.0] | 6.9 [5.9, 8.4] | 0.18 [0.14, 0.19] (24) |
| Borda | 9/9 | 1.0 [1.0, 2.0] | 5.6 [0.7, 6.2] | 0.20 [0.09, 0.35] (9) | 40/40 | 2.0 [2.0, 2.0] | 11.7 [9.5, 12.6] | 0.27 [0.25, 0.31] (40) |
| Condorcet | 9/9 | 3.0 [2.0, 4.0] | 10.0 [6.7, 16.7] | 0.59 [0.41, 1.04] (9) | 33/40 | 4.0 [4.0, 4.0] | 20.5 [18.0, 22.4] | 0.50 [0.47, 0.55] (33) |

| | rank 2–10 | | | | rank 11–50 | | | |
|---|---|---|---|---|---|---|---|---|
| | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ | feas. | $K$ | $L_1$ | $L_1/L_1^{\text{mean}}$ |
| mean | 9/9 | 23.0 [4.0, 32.0] | 156.3 [22.5, 222.4] | – | 6/40 | 46.0 [40.0, 51.0] | 321.2 [278.2, 351.3] | – |
| median | 9/9 | 4.0 [1.0, 10.0] | 3.3 [0.4, 16.6] | 0.04 [0.00, 0.09] (9) | 17/40 | 12.0 [10.0, 15.0] | 37.5 [31.7, 53.6] | 0.09 [0.07, 0.11] (6) |
| min | 6/9 | 1.5 [0.5, 2.5] | 2.3 [1.0, 5.3] | 0.03 [0.00, 0.09] (6) | 3/40 | 3.0 [2.0, 5.0] | 10.4 [6.2, 15.5] | 0.04 [0.02, 0.06] (2) |
| Borda | 9/9 | 20.0 [5.0, 30.0] | 54.0 [5.6, 100.5] | 0.36 [0.28, 0.94] (9) | 7/40 | 45.0 [38.0, 48.0] | 187.8 [143.1, 230.8] | 0.55 [0.51, 0.64] (6) |
| Condorcet | 9/9 | 16.0 [7.0, 22.0] | 20.7 [0.9, 57.6] | 0.16 [0.02, 0.22] (9) | 16/40 | 28.0 [27.0, 29.0] | 104.6 [90.4, 113.9] | 0.28 [0.21, 0.31] (6) |
| Copeland | 9/9 | 16.0 [7.0, 22.0] | 20.7 [0.9, 57.6] | 0.16 [0.02, 0.22] (9) | 16/40 | 28.0 [27.0, 29.0] | 104.6 [90.4, 113.9] | 0.28 [0.21, 0.31] (6) |

for the unweighted mean is specific to equal weights; these runs use targets at ranks 2–100 (OLLB) and 2–40 (HELM), not all targets.

**Which of our predictions held.** The plan fixed three hypotheses before the full results (H1–H3, Appendix D). H2 (cost rises with rank, feasibility falls as caps tighten) holds on both datasets. H1 (order-statistic and majority rules need *more* tasks but possibly fewer points than the mean) holds only in part: they need fewer points, but not more tasks, except that the Condorcet requirement touches more tasks than the mean on OLLB at $n = 6$ ($K = 4$ against 2 at ranks 11–50); on HELM the median needs fewer tasks than the mean ($K = 12$ against $46$). H3 is examined next.

**Number of tasks.** Figure 2 draws random subsets of $n \in \{3, 6, 12, 24, 48\}$ HELM subjects and takes the model ranked second or fifth within each subset as the target. For the rank-2 target $K/n$ falls with $n$ under every rule, and for the rank-5 target the mean and Borda rules keep touching about $0.4$ of the tasks while the median, minimum and plurality fall. For the rank-5 target the mean rule's $L_1/n$ stays roughly constant in $n$ while that of the median, minimum and plurality falls; for the rank-2 target all rules fall. H3 (that $K/n$ of the median and majority rules does not shrink) is therefore not supported for the median. The bands are wide (60 subset draws), so we make no claim about differences between neighbouring $n$. OLLB, with $n = 6$, is a single small-$n$ point and supports no scaling claim.

**Robustness.** Figure 3 resamples evaluation noise (binomial standard errors from item counts; 300 draws on OLLB, 200 on HELM) and tasks (bootstrap over tasks, same caps; 300 and 200 draws), for the mean, median, minimum, Borda and Condorcet rules; the exact rank-rule programs on OLLB use 60 draws and the strata targets only. In every case where the observed target is feasible and has a draw interval, the observed $L_1$ lies inside the 2.5–97.5 percentile interval (OLLB noise 30 of 30 target–rule cases, OLLB bootstrap 30 of 30, HELM noise 21 of 21, HELM bootstrap 21 of 21), and mostly slightly below the draw median (29/30, 23/30, 15/21, 18/21); the observed cost is thus not a departure from the noise interval. The intervals are narrow for the mean on OLLB beyond rank 10, and wide for the median and minimum, for which a large share of resampled leaderboards make the target infeasible. Feasibility at the frontier is noise-sensitive: the HELM rank-20 target is infeasible under the mean on the observed data yet feasible in a minority of draws, and the rank-40 target is infeasible in every draw. A target whose gap to first place is within the evaluation noise therefore cannot be said to need any particular investment: a win by the margin $\varepsilon$ would not survive resampling.

## 6 Discussion and limitations

**What the results say.** Under caps taken from observed fine-tuning gains, a developer who can lift only the target's own scores, and whose rivals stand still, tops these leaderboards only from near the top, and how near depends on the rule: the median reaches further down than the mean on both

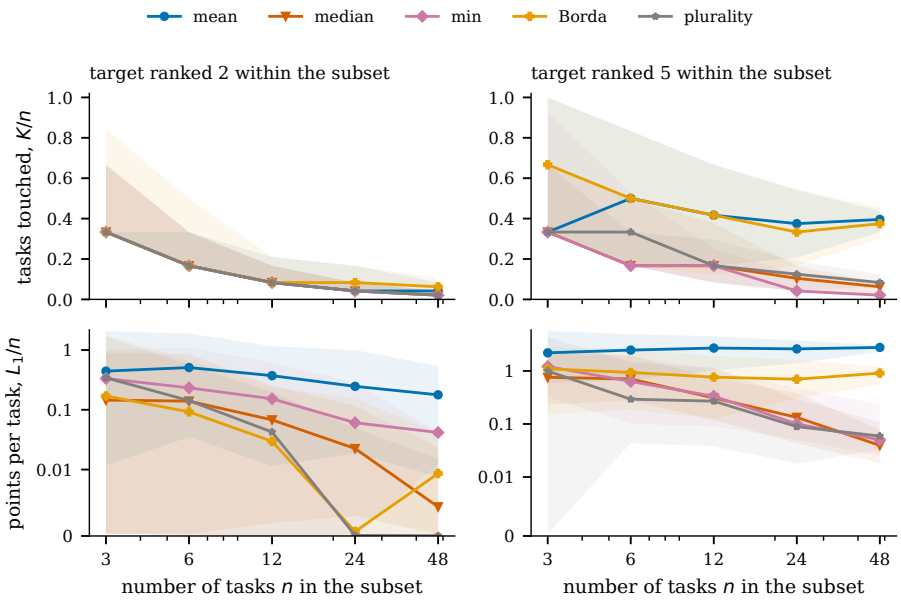



Figure 2: HELM: investment versus the number of tasks $n$ in a random subset of subjects, for the target ranked second (left) or fifth (right) within the subset under the mean; bands span the 2.5–97.5 percentiles over 60 subset draws, among feasible draws, cap `emp99T`. Generated by `figures/src/fig_scaling.py`.

datasets, and the minimum is rarely feasible. Where it is feasible the mean rule demands the most points, because it must close $n$ times the gap (Proposition 1) whatever the spread, while the median and minimum pay only for a majority of tasks or for the weak tasks. Among the score rules the mean is thus the hardest to manipulate, in points and in the rank that can be reached; the median is cheap in points and reaches further, and the minimum is cheap when feasible but needs every weak task to be raised and so is limited by the cap. Borda and the Condorcet requirement lie between the mean and the median in points, and on HELM Borda reaches only one rank further than the mean (rank 17 against 16). These orderings come with caveats we state next.

**Conditions under which the conclusions fail or are not established.** (1) *The cap.* Feasibility is determined by the cap, not by the rule, far from the top (oracle rows of Table 1). The OLLB cap uses gains of published fine-tunes over their declared base models; this is selection-biased toward successes and says nothing about what a determined developer could achieve with a new data source. The HELM cap is a transfer of the OLLB MMLU-PRO gain to 57 subjects and has not been measured on HELM. Headline claims are therefore stated per dataset. (2) *Independence across tasks.* We let each task be raised at no cost to others; targeted training can trade off tasks, which would make the true cost higher. (3) *Fixed rivals.* Other developers do not respond; leaderboards with many strategic developers need a game model such as that of Chen et al. [2026]. (4) *Score tables, not training.* The investment is in score points, not compute, data or money, so "cheap" means few score points, not little effort. We make no causal claim that any such score change can be produced by real training. (5) *Ranks within noise.* The margin of $10^{-3}$ points that defines "first" is below the evaluation noise of the scores; a win by that margin would not survive resampling (Figure 3), and the intervals for the median and minimum are wide. (6) *Rule coverage and size.* Rank rules were solved exactly for the 50 best OLLB targets only, so their feasibility fractions there cannot be compared with those of the score rules; the Condorcet requirement is a sufficient condition for Condorcet-consistent rules and therefore an upper bound for Copeland-type rules (on HELM the exact Copeland problem has the same feasible targets as the Condorcet requirement, 25 of 78 under `emp99T`, and the same median cost on the jointly feasible targets; Copeland is not solved on the full OLLB); Kemeny and Schulze were not studied. (7) *Small $n$ and small $M$.* OLLB has six tasks and HELM 79 models, so rank bins on HELM hold few targets and their intervals are wide; scaling in $n$ comes from HELM subsets only and is a statement about random subsets of one benchmark family. (8) *Data snapshot.* Both leaderboards are frozen snapshots (OLLB 2025-03, HELM v1.13.0); we do not claim anything about

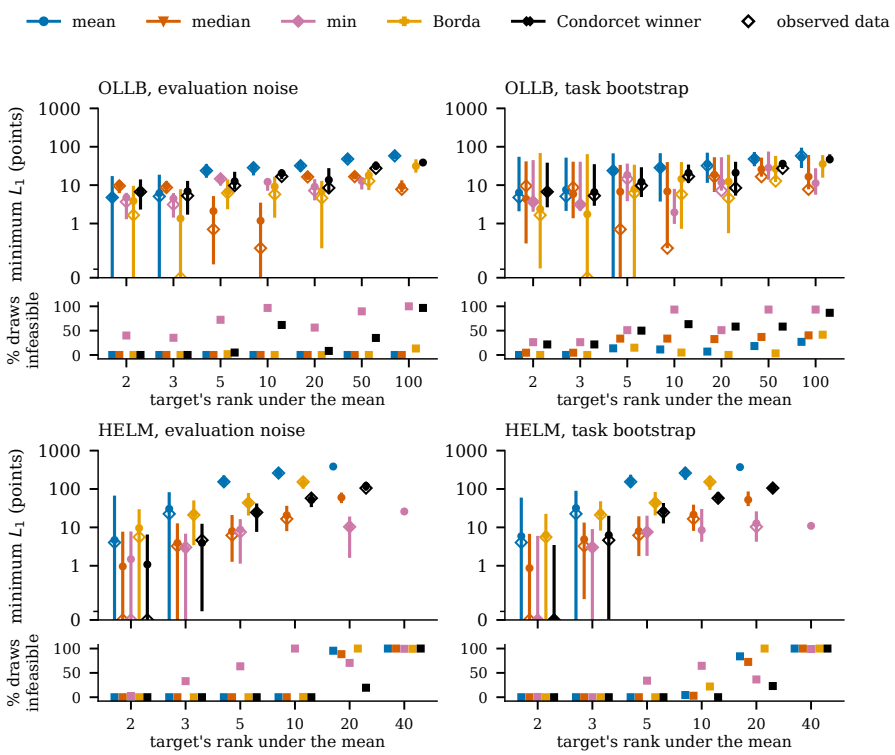



Figure 3: Stability of the minimum total gain $L_1$ (upper part of each panel) and percentage of resampled leaderboards in which the target cannot become first (strip below) under evaluation noise (left) and task bootstrap (right) on OLLB (cap `emp99`; ranks 2–100) and HELM (cap `emp99T`; ranks 2–40), for five rules. Open diamonds: value on the observed data; dots and bars: median and 2.5–97.5 percentile range of the feasible draws (a spread of the draws, not an interval for the median). Draws: 300 for OLLB score rules, 60 for OLLB Borda and Condorcet, 200 on HELM. Generated by `figures/src/fig_robust.py`.

later models. (9) *Resampling of rank rules.* The noise and bootstrap runs for the OLLB Borda and Condorcet rules use 60 draws and the stratum targets only (ranks 2–100), so their intervals are coarser than those of the score rules.

**Broader impact.** The analysis tells leaderboard designers how exposed each rule is to targeted training under stated assumptions, which supports choosing rules and caps (for instance holding out sub-datasets) deliberately. It could also be read as a recipe for gaming a ranking. We think the risk is small: it needs no new capability beyond fine-tuning on data a developer already can use, and our results indicate that for most models it is infeasible under the observed gains. We do not evaluate any model's training.

## 7 Reproducibility

All numbers, tables and figures are produced by the supplementary code from the pinned public data (versions, checksums, seeds, commands, expected run times are in `REPRODUCE.md`; Appendix C maps every figure and table of this paper to its generating script). The solvers are deterministic; random draws use seed 0. The full pipeline needs no GPU and takes a few CPU hours on a shared two-CPU container. Neither leaderboard declares a licence for its score data; we release only derived score tables. No material about the published study that motivated this question was searched for or read, and none was encountered (reproduction notes, section on forbidden material).

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# A Proofs of Propositions 1–3

**Proposition 1.** $f(y) = \frac{1}{n}\sum_t y_t \geq \tau$ iff $\sum_t (y_t - x_t) \geq n(\tau - f(x))$, and $\sum_t (y_t - x_t) \leq \sum_t h_t$ with each $0 \leq y_t - x_t \leq h_t$; so feasibility is $\sum_t h_t \geq n(\tau - f(x))$, i.e. $\sum_t c_t \geq n\tau$, and the minimum of the (linear) total is the requirement itself, attained by any feasible spread. $K$ is minimised by filling the largest headrooms first, since touching $k$ tasks can supply at most the sum of the $k$ largest $h_t$. The uniform raise by $\tau - f(x)$ attains $L_\infty$, and no spread has a smaller maximum because the average raise is $\tau - f(x)$.

**Proposition 2.** $\min_t y_t > \tau$ requires $y_t > \tau$ for every $t$, hence $c_t > \tau$ wherever $x_t \leq \tau$ (tasks with $x_t > \tau$ need no change). The cheapest $y_t$ on a deficient task is $\tau$ (up to $\varepsilon$), so $L_1 = \sum_t (\tau - x_t)^+$, and $K$ counts the deficient tasks. Both are forced.

**Proposition 3.** For odd $n$ the median exceeds $\tau$ iff at least $m$ coordinates exceed $\tau$. Tasks with $x_t > \tau$ already count. We need $m' = (m - \#\{x_t > \tau\})^+$ more tasks. Lifting task $t$ above $\tau$ costs at least $d_t$ (and is possible iff $c_t > \tau$); costs are separable, so the cheapest set is the $m'$ smallest finite $d_t$ and $K = m'$, which cannot be smaller because each lifted task adds one to the count.

# B Further results

Table 3: Closed forms (Propositions 1–3) against the exact solvers and an independent generic LP/MILP (60-target sample), over all caps. Generated by `figures/src/tables.py` from `results/theory_check.csv`.

| data | rule | reference | max$\lvert\Delta L_1\rvert$ | max$\lvert\Delta K\rvert$ | feas. mismatches | caps |
|---|---|---|---|---|---|---|
| OLLB | mean | exact solver | 5.7e-14 | 0 | 0 | 7 |
| OLLB | mean | generic LP/MILP | 8.5e-14 | 0 | 0 | 7 |
| OLLB | min | exact solver | 0.0e+00 | 0 | 0 | 7 |
| OLLB | min | generic LP/MILP | 4.3e-14 | 0 | 0 | 7 |
| OLLB | median | exact solver | solver only | – | – | 7 |
| HELM | mean | exact solver | 4.5e-13 | 0 | 0 | 8 |
| HELM | mean | generic LP/MILP | 1.8e-12 | 0 | 0 | 8 |
| HELM | min | exact solver | 0.0e+00 | 0 | 0 | 8 |
| HELM | min | generic LP/MILP | 1.3e-12 | 0 | 0 | 8 |
| HELM | median | exact solver | 5.7e-14 | 0 | 0 | 8 |
| HELM | median | generic LP/MILP | 1.5e-12 | 0 | 0 | 8 |

Table 4: Paired comparison with the mean rule at the primary caps: median ratio $L_1(\text{rule})/L_1(\text{mean})$ over targets feasible under both, with 95% bootstrap interval over targets; second line: targets feasible under both / under at least one of the two rules. "–": fewer than 3 targets feasible under both. Top: OLLB (`emp99`); † the 50 best targets only. Bottom: HELM (`emp99T`). Strata are rank under the mean. Generated by `figures/src/tables.py`.

| rule vs. mean | all | rank 2-5 | rank 6-10 | rank 11-30 | rank 31-100 |
|---|---|---|---|---|---|
| median | 0.39 [0.38, 0.40]<br>435/561 | 1.38 [0.03, 2.00]<br>4/4 | 0.02 [0.00, 0.18]<br>4/5 | 0.13 [0.09, 0.29]<br>20/20 | 0.37 [0.32, 0.40]<br>69/70 |
| min | 0.11 [0.10, 0.11]<br>237/581 | 0.61 [0.35, 0.77]<br>4/4 | –<br>1/5 | 0.21 [0.10, 0.28]<br>12/20 | 0.14 [0.10, 0.18]<br>29/70 |
| Borda† | 0.27 [0.25, 0.30]<br>48/546 | 0.26 [0.09, 0.35]<br>3/4 | 0.20 [0.18, 0.36]<br>5/5 | 0.27 [0.23, 0.37]<br>20/20 | 0.27 [0.25, 0.31]<br>20/70 |
| Condorcet† | 0.52 [0.47, 0.57]<br>42/546 | 0.91 [0.41, 1.41]<br>4/4 | 0.59 [0.40, 0.59]<br>5/5 | 0.51 [0.37, 0.56]<br>16/20 | 0.49 [0.47, 0.57]<br>17/70 |
| plurality† | 0.66 [0.59, 0.86]<br>36/546 | 0.84 [0.64, 1.09]<br>4/4 | 0.75 [0.62, 1.30]<br>5/5 | 0.60 [0.49, 1.00]<br>18/20 | 0.68 [0.52, 0.93]<br>9/70 |

| rule vs. mean | all | rank 2-3 | rank 4-5 | rank 6-10 | rank 11-20 |
|---|---|---|---|---|---|
| median | 0.07 [0.04, 0.10]<br>15/26 | –<br>2/2 | –<br>2/2 | 0.06 [0.00, 0.09]<br>5/5 | 0.09 [0.07, 0.11]<br>6/9 |
| min | 0.05 [0.02, 0.06]<br>7/16 | –<br>1/2 | –<br>2/2 | –<br>2/5 | –<br>2/7 |
| Borda | 0.51 [0.32, 0.59]<br>15/16 | –<br>2/2 | –<br>2/2 | 0.36 [0.32, 0.59]<br>5/5 | 0.55 [0.51, 0.64]<br>6/7 |
| Condorcet | 0.21 [0.12, 0.28]<br>15/25 | –<br>2/2 | –<br>2/2 | 0.19 [0.09, 0.28]<br>5/5 | 0.28 [0.21, 0.31]<br>6/10 |
| Copeland | 0.21 [0.12, 0.28]<br>15/25 | –<br>2/2 | –<br>2/2 | 0.19 [0.09, 0.28]<br>5/5 | 0.28 [0.21, 0.31]<br>6/10 |
| plurality | 0.05 [0.03, 0.08]<br>14/45 | –<br>2/2 | –<br>1/2 | 0.05 [0.02, 0.08]<br>5/5 | 0.07 [0.04, 0.08]<br>6/10 |

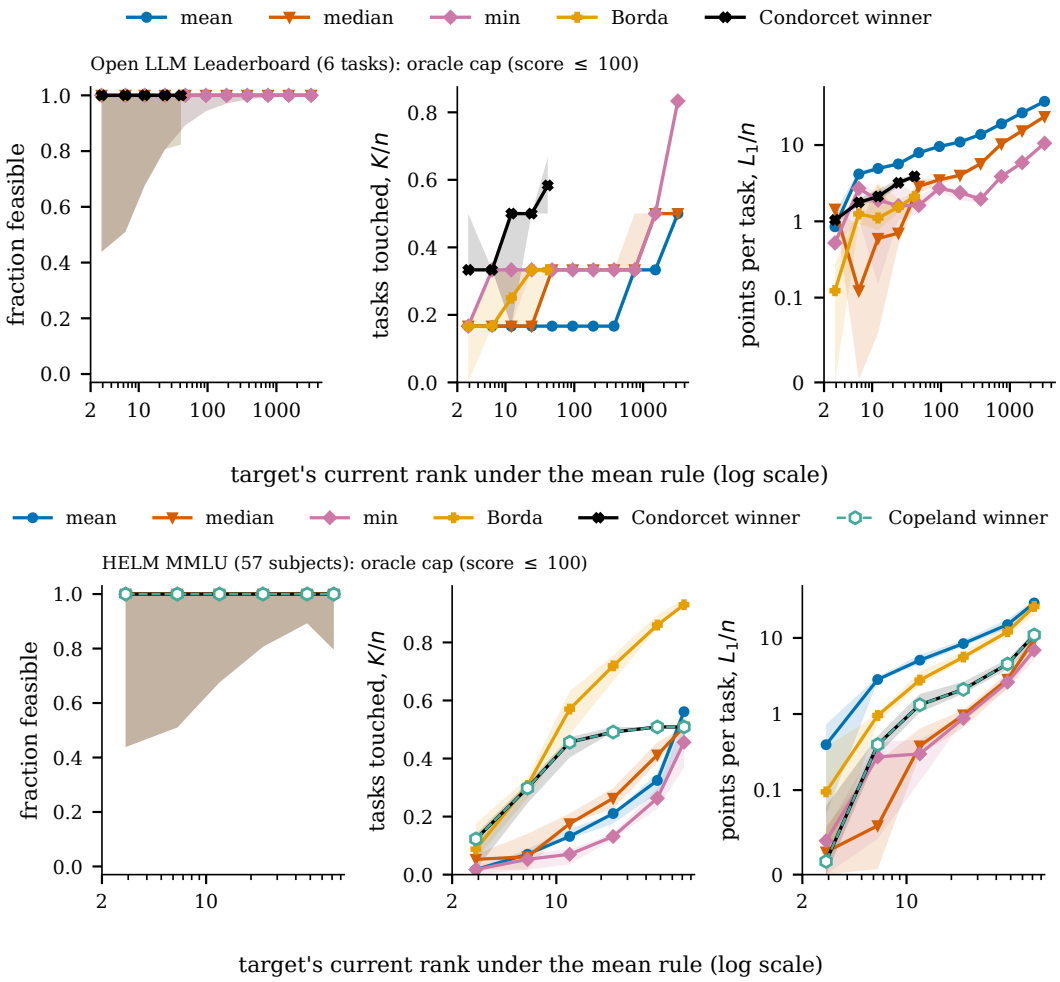



Figure 4: As Figure 1 with the oracle cap `ceil` (scores up to 100): a lower bound on any required investment; every target is feasible, so the differences between rules are not caused by the cap.

## C Where each figure and table comes from

Commands are run from the repository root; `R` is `results/`. Every figure has a `.json` sidecar and every table set a `tables/MANIFEST.json` with the SHA-256 of each input file (see `figures/MANIFEST.md`).

## D Hypotheses fixed before the full results

The plan fixed three hypotheses from the closed forms before the full result set was available. H1: with loose caps, score-sum rules need the same $L_1$ regardless of $K$, while order-statistic and majority rules need more tasks but possibly fewer points than the mean. H2: required investment grows with the target's initial rank; feasibility falls as caps tighten; rule ordering can change with the cap. H3: for fixed per-task caps the number of tasks that majority and median rules need does not shrink with $n$, while the mean rule's $L_1/n$ depends on the density of gaps. Anything beyond these is exploratory.

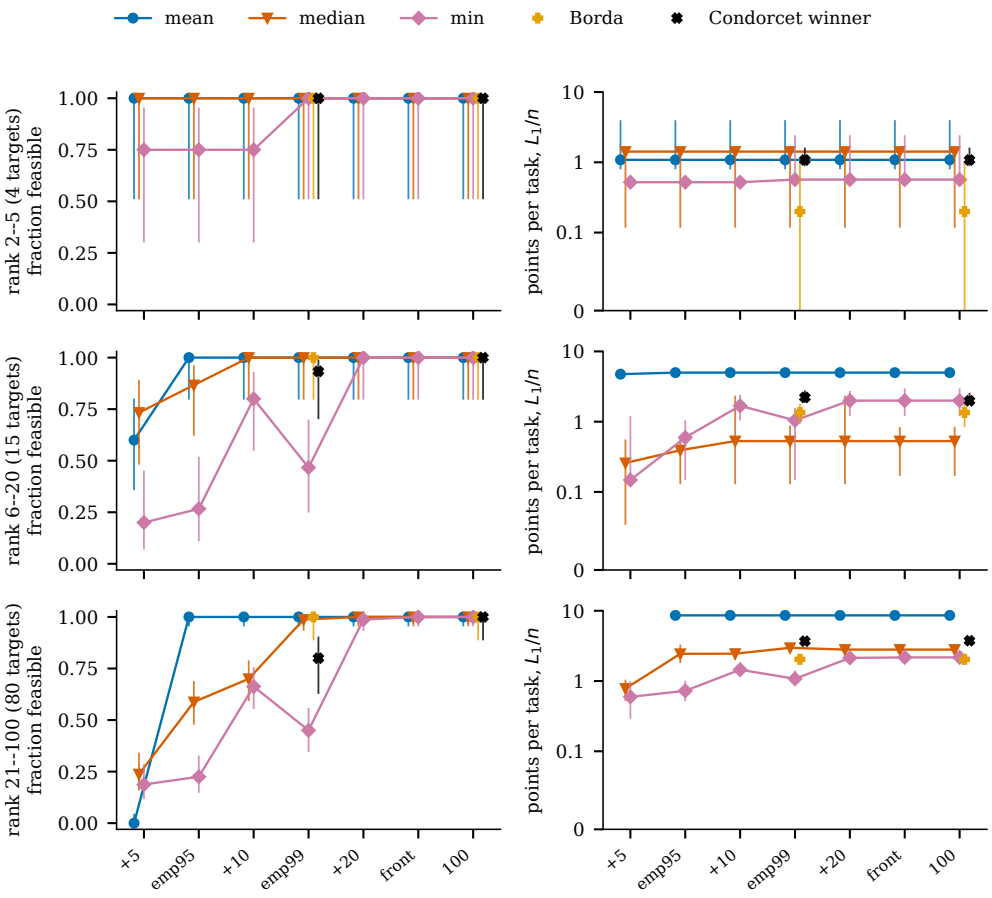



Figure 5: Effect of the cap on feasibility (left) and on the median $L_1/n$ among feasible targets (right) for three rank strata under the mean, for OLLB (7 caps); the HELM version is Figure 6. Caps: $+h$ = at most $h$ points per task; `emp`: per-task empirical percentile of fine-tune gains; `front`: best rival's score; 100: the ceiling. The caps are ordered by the share of targets feasible, averaged over mean, median and minimum, so the order is data-derived, not a cap size, and differs between the datasets. For the mean rule the cost among feasible targets does not depend on the cap (Proposition 1); only the feasible fraction does. Median cost is omitted where fewer than 3 targets are feasible; OLLB Borda and Condorcet (50 best targets) appear at `emp99` and the oracle cap only. HELM caps other than `ceil` and `emp99T` have no empirical HELM basis. Bars: 95% bootstrap (Wilson for fractions) intervals over targets. Generated by `figures/src/fig_cap_grid.py`.

# NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: Abstract and introduction state the two dataset-specific findings, the cap assumption and their scope; each is backed by Table 2, Table 1 and Figure 1, and Section 6 lists where they fail.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.

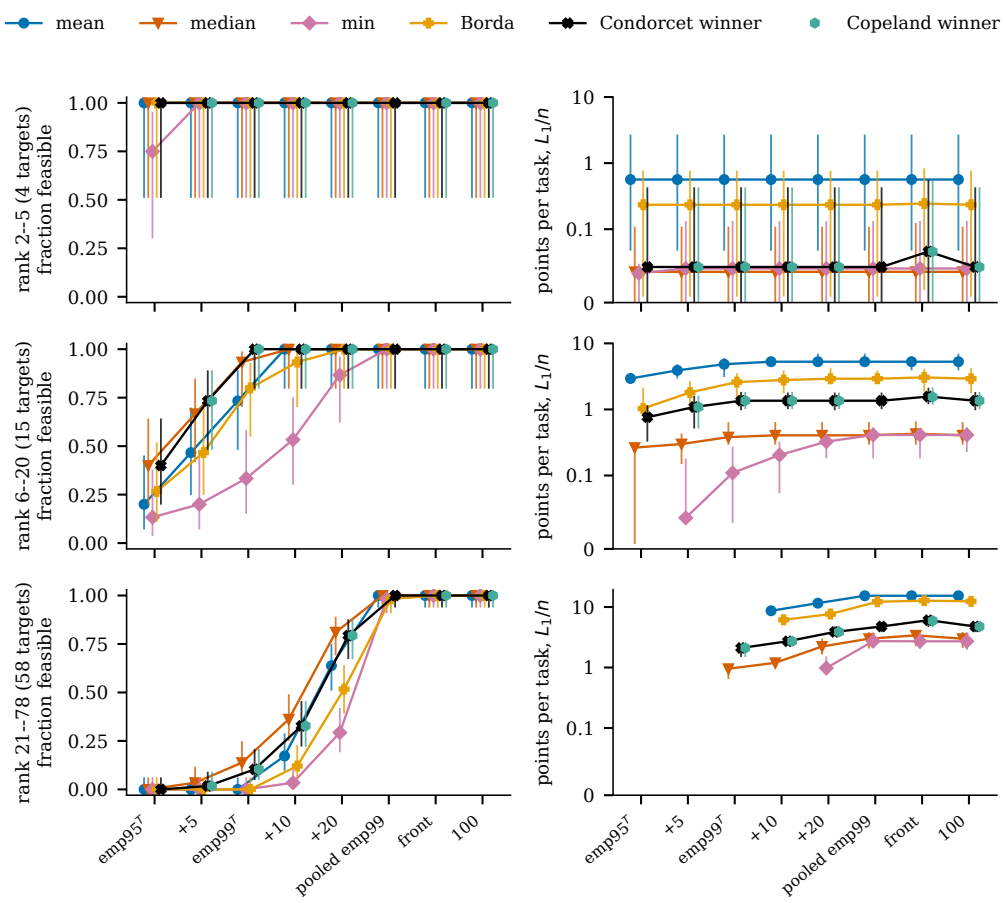



Figure 6: As Figure 5 for HELM MMLU (8 caps, including the transferred `emp95T` and `emp99T` and the pooled cap `emp99P`; all rules, including Copeland, are connected). The caps other than `ceil` and `emp99T` have no empirical HELM basis, and the caps are ordered by the feasible share on HELM, which differs from the OLLB order. Generated by `figures/src/fig_cap_grid.py`.

- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 6 (Discussion and limitations) lists cap selection bias and transfer, task independence, fixed rivals, noise, rule coverage, snapshots and the coarser resampling of the OLLB rank rules.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.

Table 5: Sensitivity of the conclusions, primary caps. *feas.*: share of targets feasible under the variant; *feas. (base)*: the same share in the baseline; median $L_1$ among feasible targets; last column: median per-target ratio of the variant's cost to the baseline cost over targets feasible under both with positive cost (their number in parentheses). Margin: points by which the target must lead; raw scores use the cap `gain10` (+10 points) because the empirical caps are defined on normalised scores. Random task weights: weighted mean with Dirichlet weights; $n$ counts target–weight draws, the baseline is the uniform-weight cost of the same target, and targets are those of ranks 2–100 (OLLB) or 2–40 (HELM). Costs are conditional on feasibility. Generated by `figures/src/tables.py`.

| variant | cap | rule | $n$ | feas. | feas. (base) | median $L_1$ | $L_1$ / baseline ($n_{\text{both}}$) |
|---|---|---|---|---|---|---|---|
| HELM: margin 1 pt | `emp99T` | mean | 78 | 0.15 | 0.19 | 229.1 | 1.33 (12) |
| HELM: margin 1 pt | `emp99T` | median | 78 | 0.28 | 0.33 | 30.1 | 1.55 (22) |
| HELM: margin 1 pt | `emp99T` | min | 78 | 0.12 | 0.12 | 4.7 | 1.50 (8) |
| HELM: margin 2 pt | `emp99T` | mean | 78 | 0.12 | 0.19 | 270.3 | 1.73 (9) |
| HELM: margin 2 pt | `emp99T` | median | 78 | 0.21 | 0.33 | 38.8 | 2.34 (16) |
| HELM: margin 2 pt | `emp99T` | min | 78 | 0.09 | 0.12 | 6.7 | 2.26 (6) |
| HELM: random task weights | `emp99T` | Dirichlet(1) | 3000 | 0.68 | 0.67 | 33.6 | 0.43 (1739) |
| HELM: random task weights | `emp99T` | Dirichlet(5) | 3000 | 0.67 | 0.67 | 74.9 | 0.68 (1836) |
| OLLB: margin 1 pt | `emp99` | mean | 4495 | 0.11 | 0.12 | 77.1 | 1.08 (500) |
| OLLB: margin 1 pt | `emp99` | median | 4495 | 0.09 | 0.10 | 29.5 | 1.15 (403) |
| OLLB: margin 1 pt | `emp99` | min | 4495 | 0.04 | 0.06 | 9.6 | 1.25 (191) |
| OLLB: margin 2 pt | `emp99` | mean | 4495 | 0.10 | 0.12 | 82.0 | 1.17 (466) |
| OLLB: margin 2 pt | `emp99` | median | 4495 | 0.08 | 0.10 | 32.2 | 1.34 (341) |
| OLLB: margin 2 pt | `emp99` | min | 4495 | 0.03 | 0.06 | 10.9 | 1.53 (115) |
| OLLB: no merged models | `emp99` | mean | 3788 | 0.11 | 0.11 | 74.1 | 1.00 (435) |
| OLLB: no merged models | `emp99` | median | 3788 | 0.09 | 0.09 | 26.6 | 1.00 (356) |
| OLLB: no merged models | `emp99` | min | 3788 | 0.07 | 0.06 | 7.4 | 0.81 (212) |
| OLLB: raw scores | `gain10` | mean | 4495 | 0.07 | 0.03 | 48.5 | 0.74 (113) |
| OLLB: raw scores | `gain10` | median | 4495 | $<$0.01 | 0.05 | 15.7 | 5.65 (20) |
| OLLB: raw scores | `gain10` | min | 4495 | 0.05 | 0.10 | 12.6 | 1.50 (217) |
| OLLB: random task weights | `emp99` | Dirichlet(1) | 3500 | 0.91 | 1.00 | 12.2 | 0.54 (3128) |
| OLLB: random task weights | `emp99` | Dirichlet(5) | 3500 | 0.99 | 1.00 | 19.1 | 0.70 (3440) |

| paper item | file(s) in `figures/`, `tables/` | script and inputs |
|---|---|---|
| Figure 1 | `F_ollb_primary`, `F_helm_primary` | `figures/src/fig_rank_rules.py R <ollb\|helm> <stem> -caps primary`; `core_*.csv` |
| Figure 2 | `F_helm_scaling` | `figures/src/fig_scaling.py R <stem>`; `summary/summary_rob_helm_tasksub_emp99T.csv` |
| Figure 3 | `F_robust` | `figures/src/fig_robust.py R <stem>`; `summary/summary_rob_*.csv`, `core_*.csv` |
| Figure 4 | `F_ollb_oracle`, `F_helm_oracle` | `fig_rank_rules.py` with `-caps ceil` |
| Figures 5, 6 | `F_ollb_capgrid`, `F_helm_capgrid` | `figures/src/fig_cap_grid.py R <ollb\|helm> <stem>`; `core_*.csv` |
| Table 1 | `T4_frontier_*` | `figures/src/tables.py R tables`; `core_*.csv` |
| Table 2 | `T3_rules_*` | same; `core_*.csv` |
| Table 3 | `T2_theory_check` | same; `theory_check.csv` |
| Table 4 | `T5_paired_*` | same; `summary/paired_*.csv` |
| Table 5 | `T6_variants_compact` | same; `summary/summary_variants.csv` |

Table 6: Generators of the figures and tables of this paper. The result files themselves are produced by `code/run_all.sh` (see `REPRODUCE.md`).

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